VIT Vidyalankar Institute of Technology Accredited A+ by NAAC	Department of Computer Engineering A.Y-2026-27

Semester	B.E. Semester VII – CMPN
Subject	DevOps
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Experiment Number	3
Experiment Title	Luggage Complaint Portal: Requirements, Architecture and Technology Setup
Objectives (Skill Set / Knowledge Tested / Imparted)	To design the minimum application architecture, select the technology stack, build tool and deployment target, and prepare a working local setup.
Program	**1. SRS summary**

ID	Functional requirement
FR1	A passenger submits a complaint with name, email, phone, flight number, PNR, bag tag, complaint type and description.
FR2	The system validates the data, shows error messages for invalid input and accepts a file upload.
FR3	The system generates a reference number in the format LCP-YYYY-NNNNNN.
FR4	A passenger tracks the status and history of a complaint using the reference number.
FR5	A reviewer logs in securely.
FR6	A reviewer sees the list of complaints.
FR7	A reviewer approves or rejects a complaint with a comment, and the status changes.

- Non-functional: simple responsive pages, reviewer pages protected by login, a health endpoint for monitoring, portability (WAR on Tomcat, Docker, systemd) and testability (unit and Selenium tests).

2. Use-case diagram

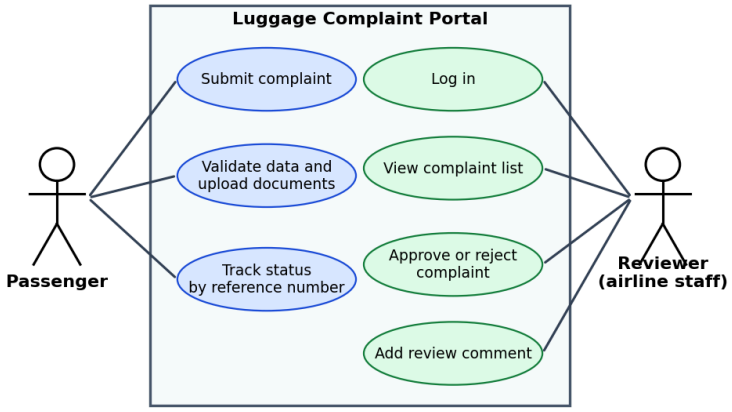


Figure: Use-case diagram (Passenger and Reviewer)

3. Architecture diagram

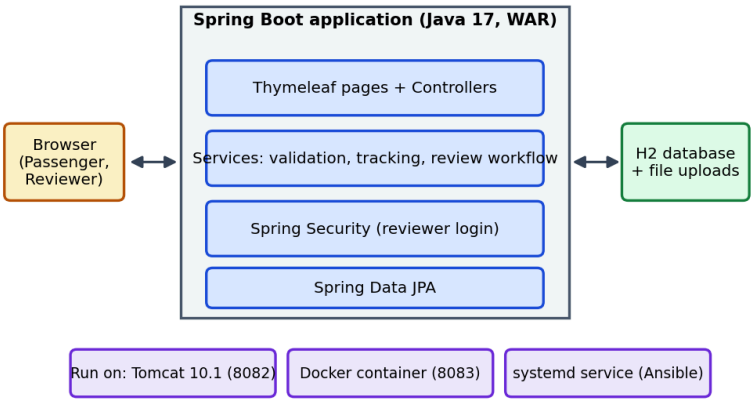


Figure: Minimum application architecture and deployment targets

4. Technology stack

Item	Choice
Language	Java 17
Framework	Spring Boot 3.3.5 with Thymeleaf, Spring Security and Spring Data JPA
Database	H2 (embedded)
Build tool	Maven (packaged as a WAR)
Deployment target	Tomcat 10.1 (also Docker and systemd)
Testing	JUnit 5 and Selenium WebDriver 4

5. Data model

Entity	Main fields

Complaint	reference number, passenger name, email, phone, flight number, PNR, bag tag, type, description, status, submitted date, attachment
Status history	when, from status, to status, changed by, note
Reviewer account	username and credentials used for the reviewer login

6. URL / API list

Method and path	Purpose
GET /	Home page with Submit, Track and Reviewer links
GET /complaints/new	Complaint form (submitted by POST from the same form)
GET /track	Track a complaint by reference number
GET /login	Reviewer login
Reviewer pages (after login)	Complaint list and approve/reject decision form
GET /health	Plain health check (returns OK)
GET /actuator/health, /actuator/info	Spring Boot health and build information

7. Working local setup

```
cd luggage-complaint-portal
mvn spring-boot:run
# open http://localhost:8081

Tomcat started on port 8081 (http) with context path '/'
Started LuggagePortalApplication in 5.996 seconds
```

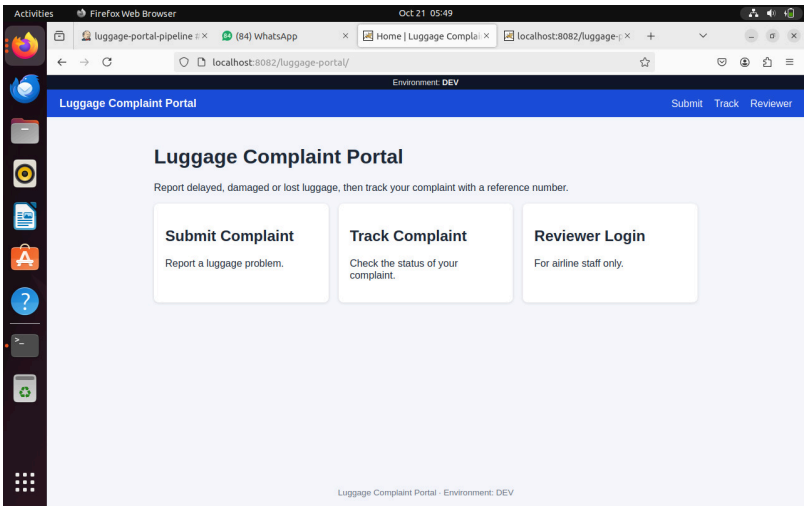


Figure: Application running (home page with environment banner)

Conclusion	The minimum architecture, technology stack, data model and URLs were defined, and the application was set up and run locally.